

Worry is associated with excessive model-based revaluation

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Introduction

- Worry has so far no empirical relation to MB planning usage.
- Clinicians observe the biased over-use of planning in worry.
- Past attempts have not considered RT or used tasks that require longer offline revaluation
- In a large initial study, we observe worriers spend more time deliberating without a benefit in retrospective revaluation tasks.
- We sought to replicate and model this effect in the present study

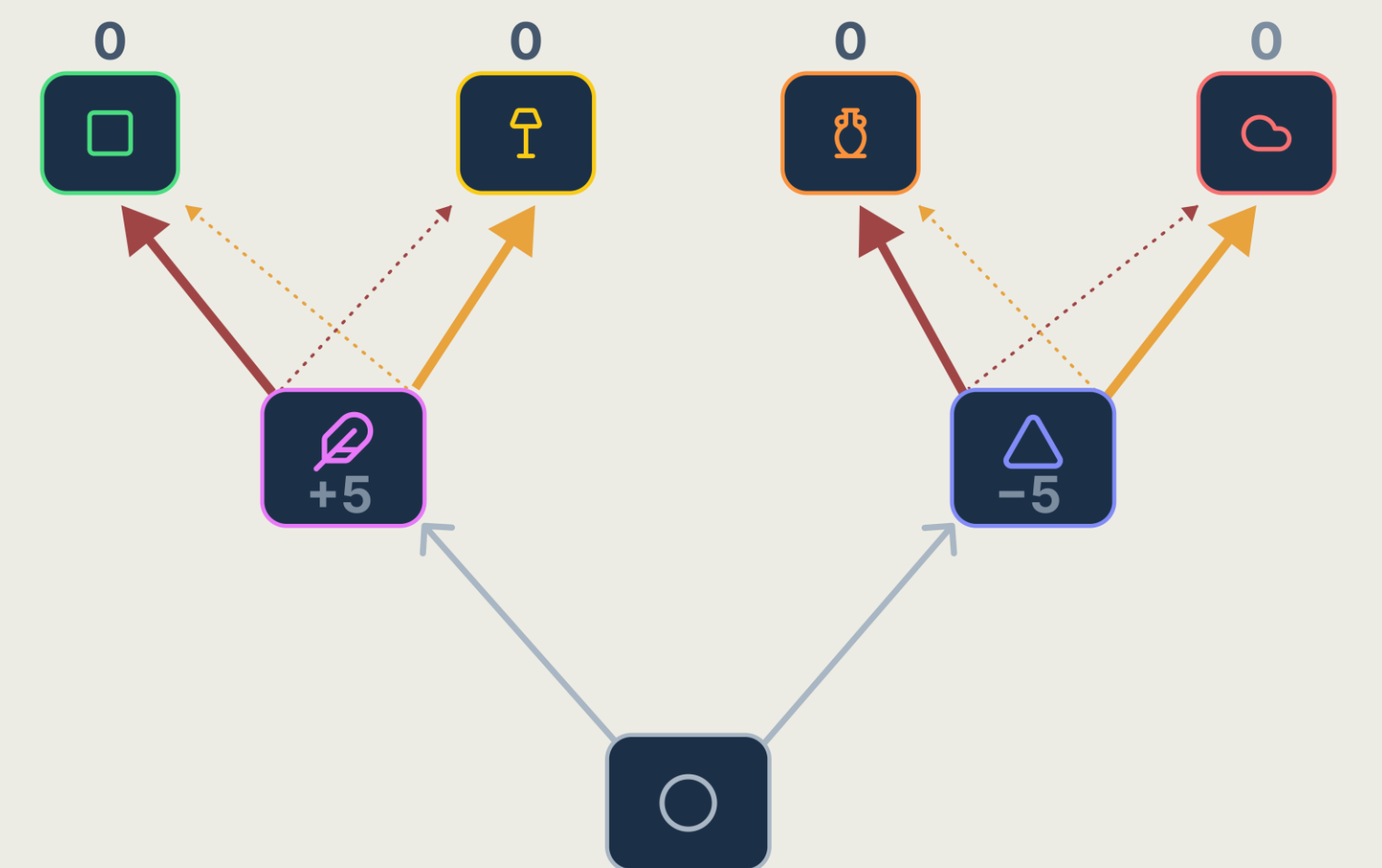
Hypotheses

- In planners: higher somatic anxiety → more overthinking
- In planners: higher trait worry → more overthinking
- Planners deliberate longer per trial than non-planners

Task

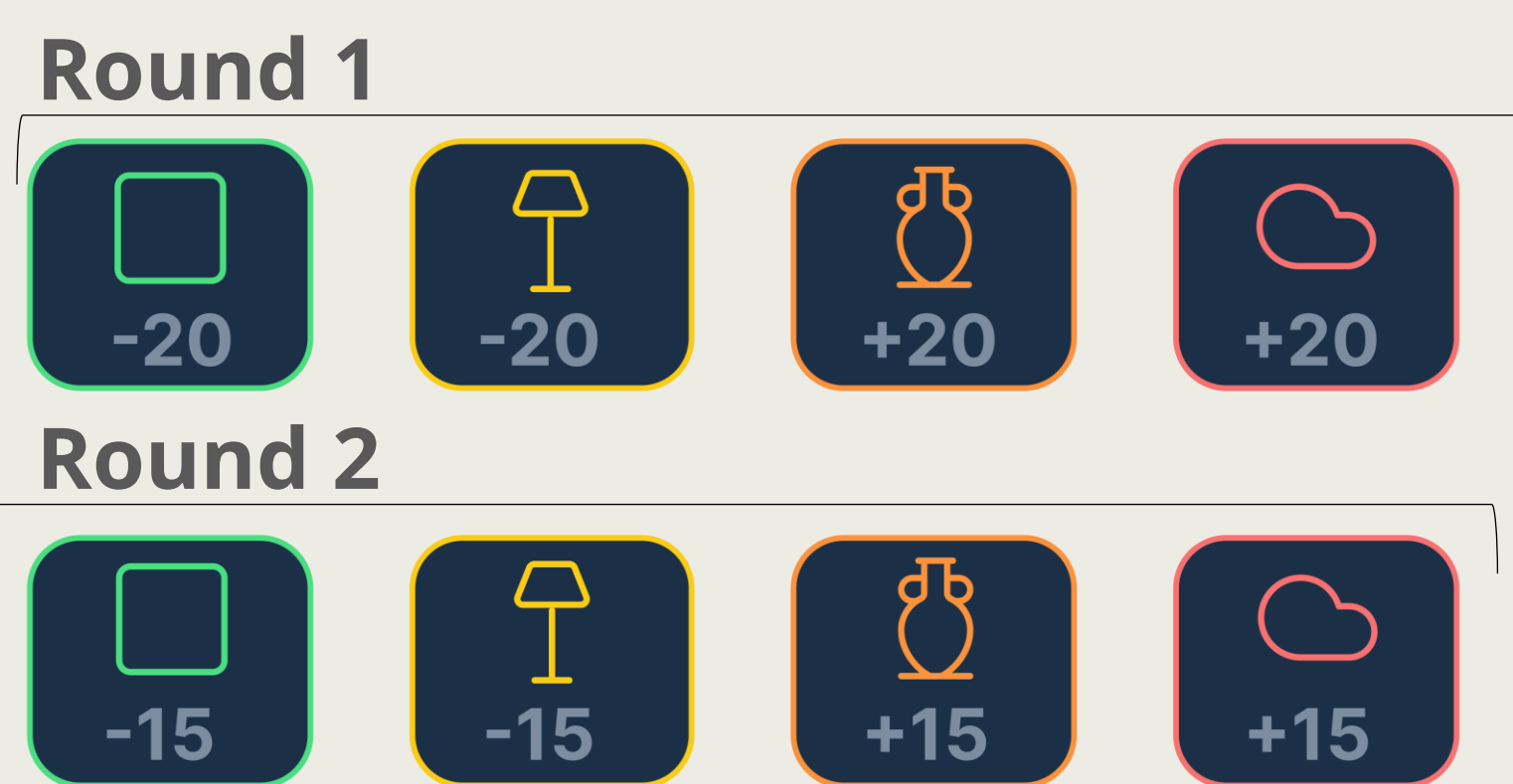
1 Structure and value learning

- First – Force choice structure learning
- Second, free-choice (except for initial choices) value learning

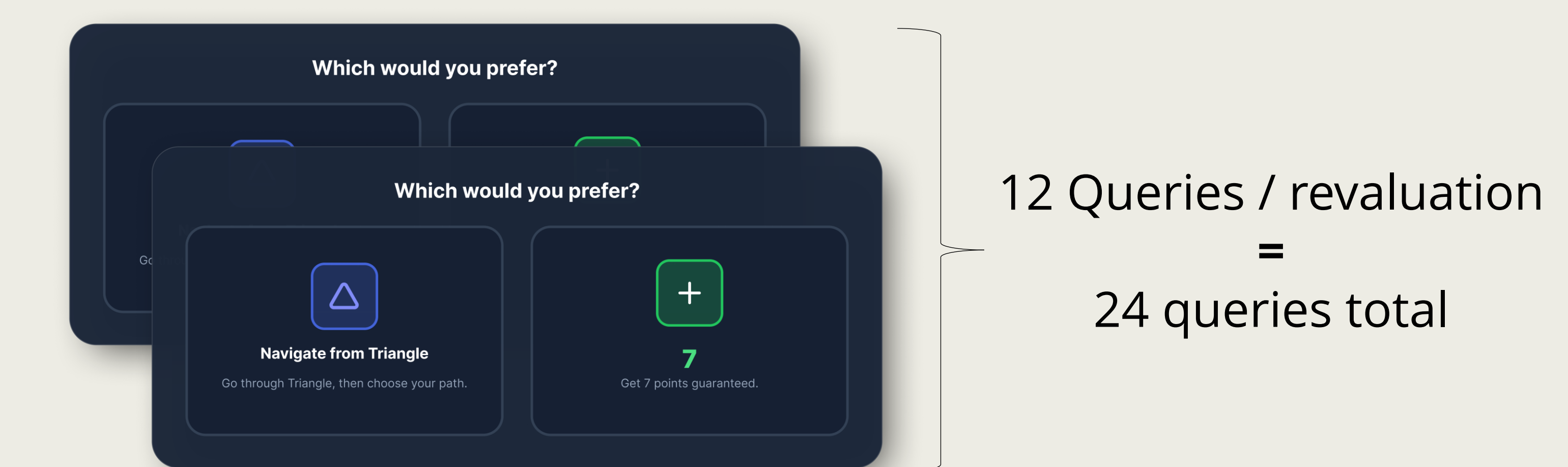


2 Instructed revaluation phase (twice)

Two revaluation phases instructs changes to end-states



3 Queries: Measuring deliberation time & planning

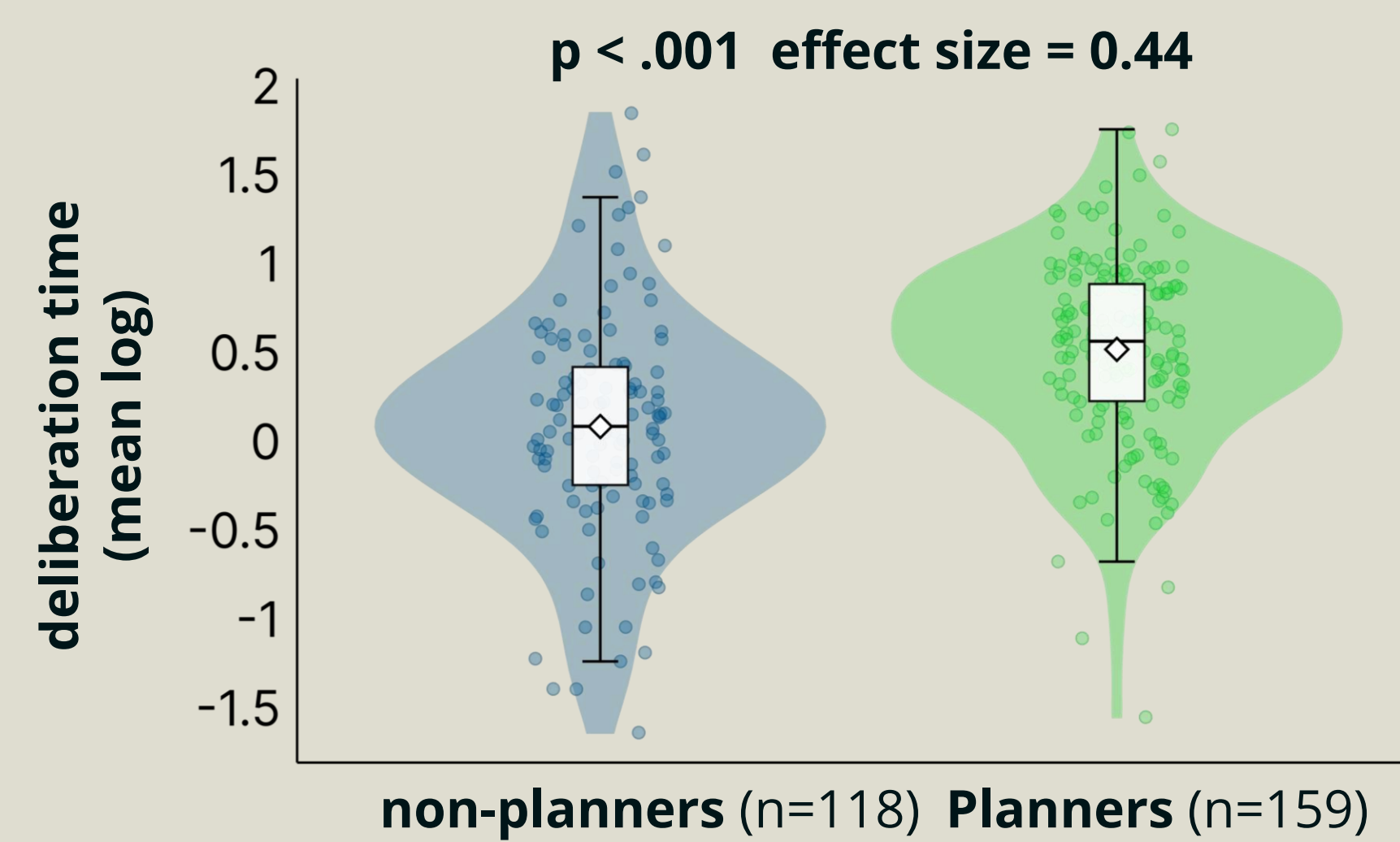


Participants & Measures

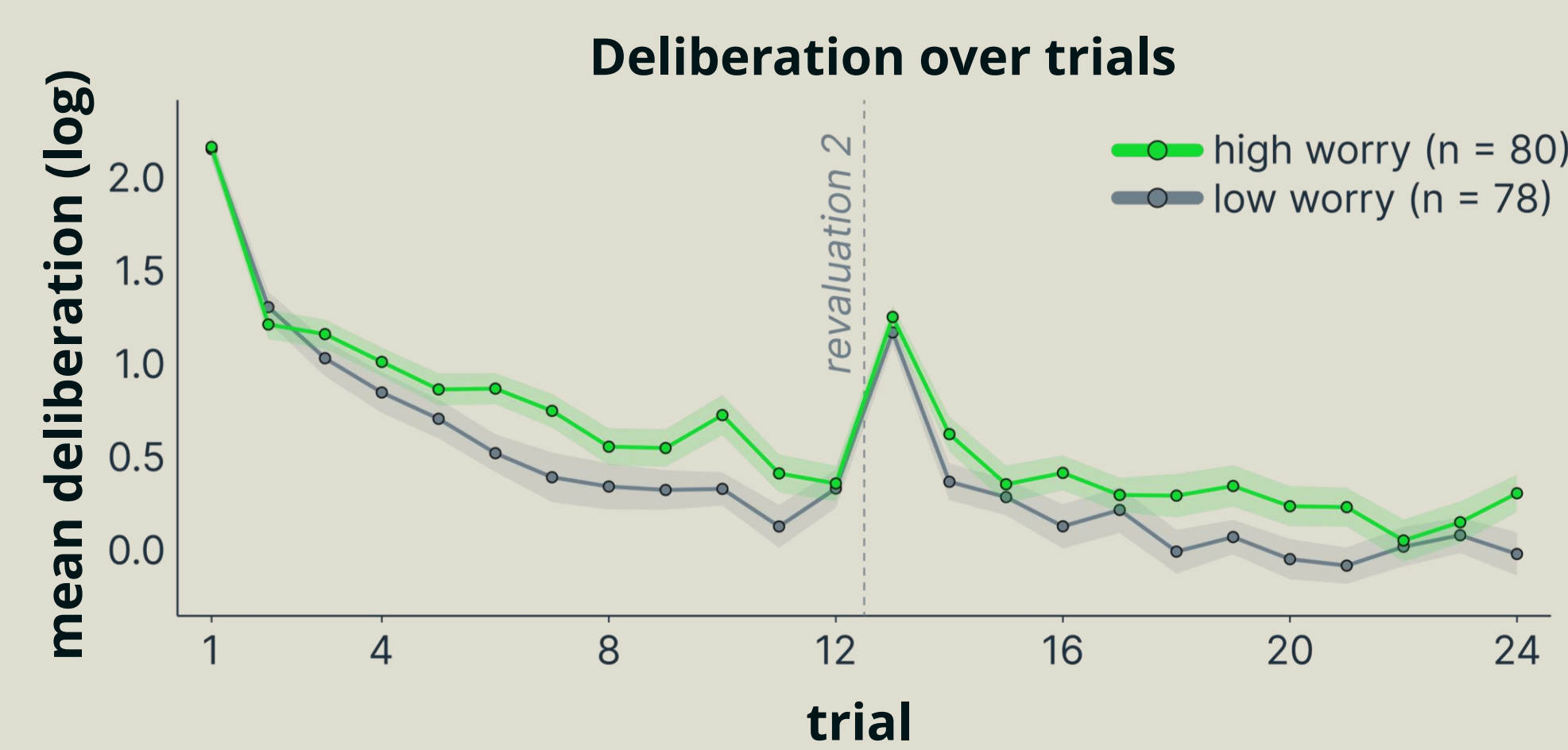
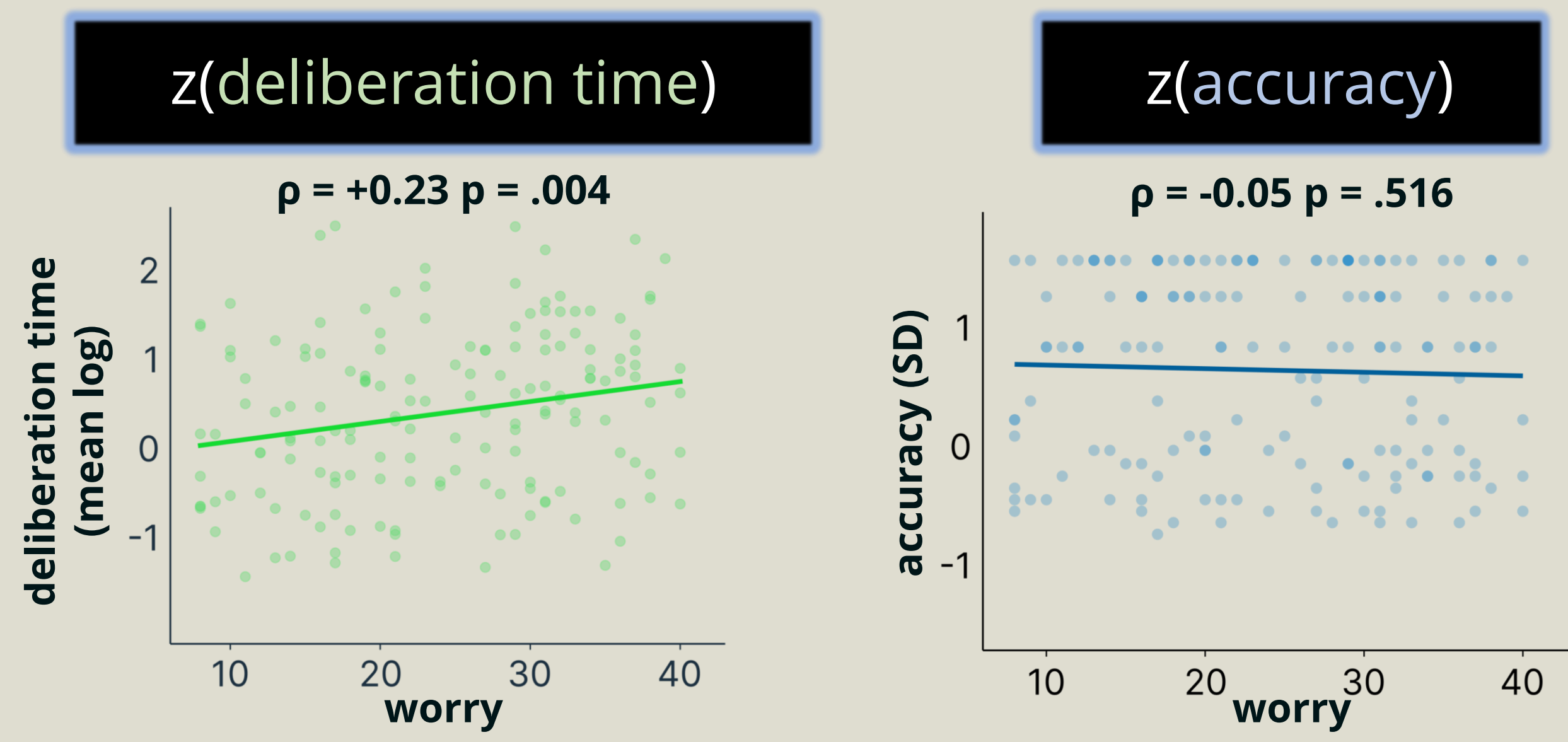
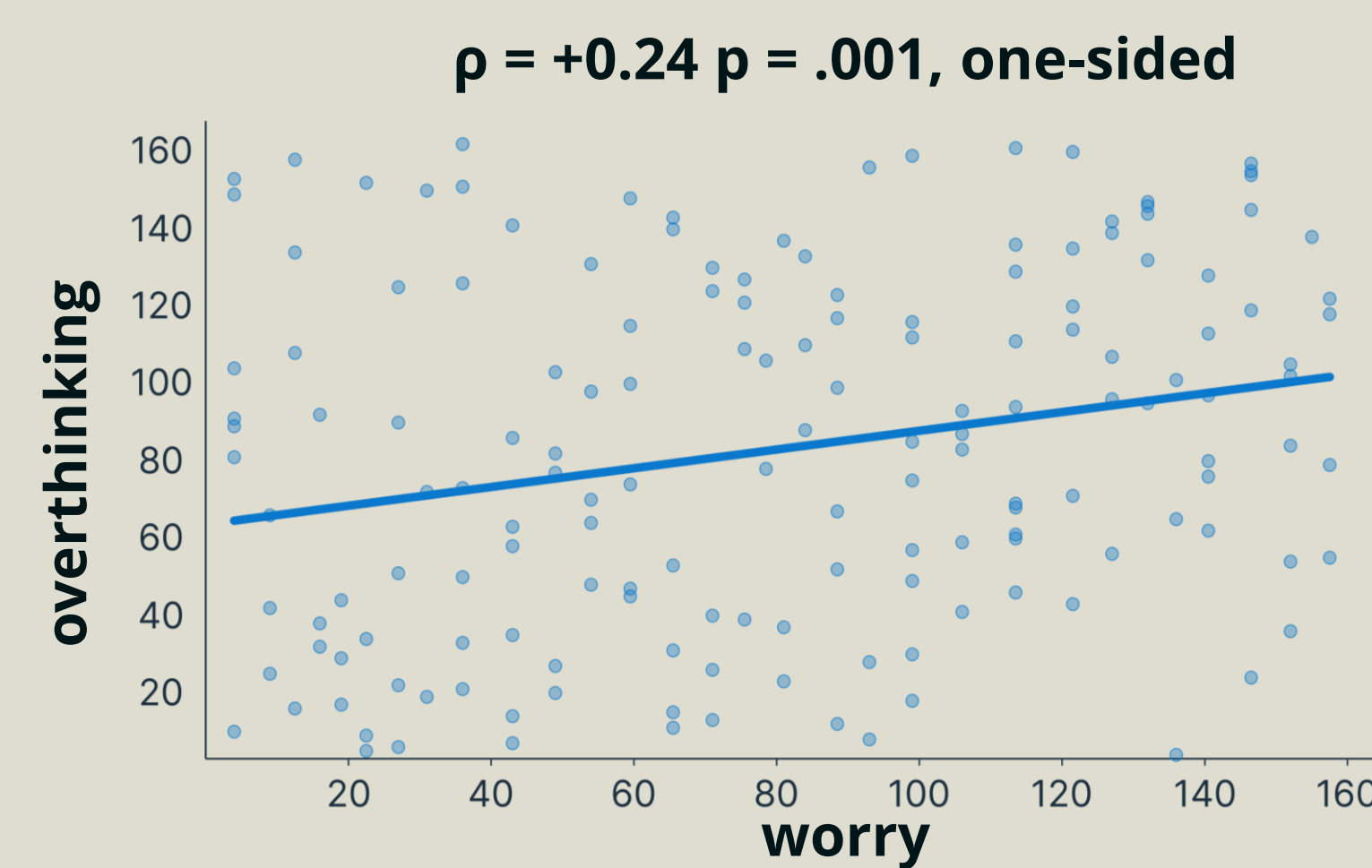
- 277 adults (age M = 34.5; 51% female, 49% male) recruited via Prolific
- Worry: PSWQ-A³, somatic anxiety: MASQ arousal scale⁴.
- 159 model-based "planners" and 118 model-free responders.
- At least 50% accuracy according to MB revaluation on either stimuli

Descriptive Results

Distinguishing MB and MF via deliberation time



Overthinking = z(deliberation time) – z(accuracy)

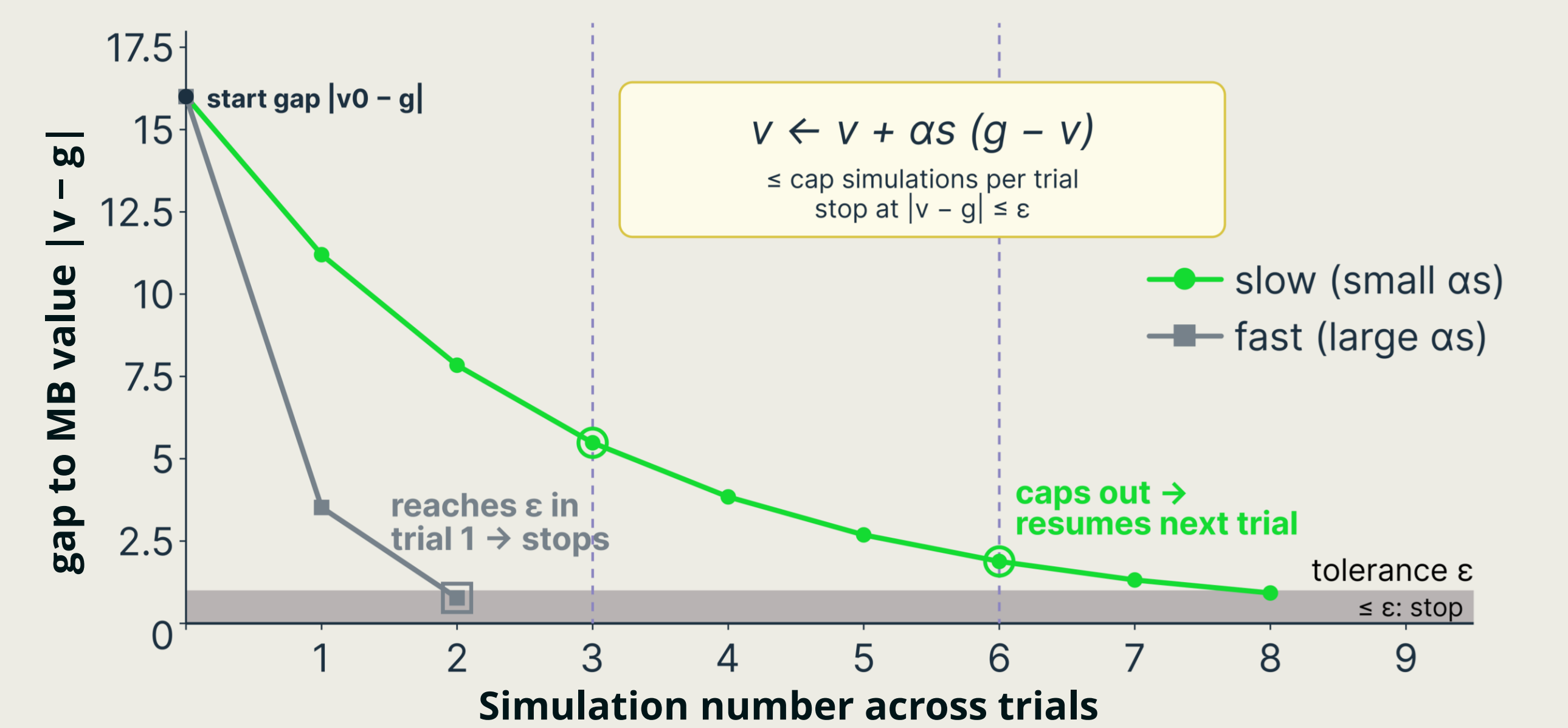


Modelling Results

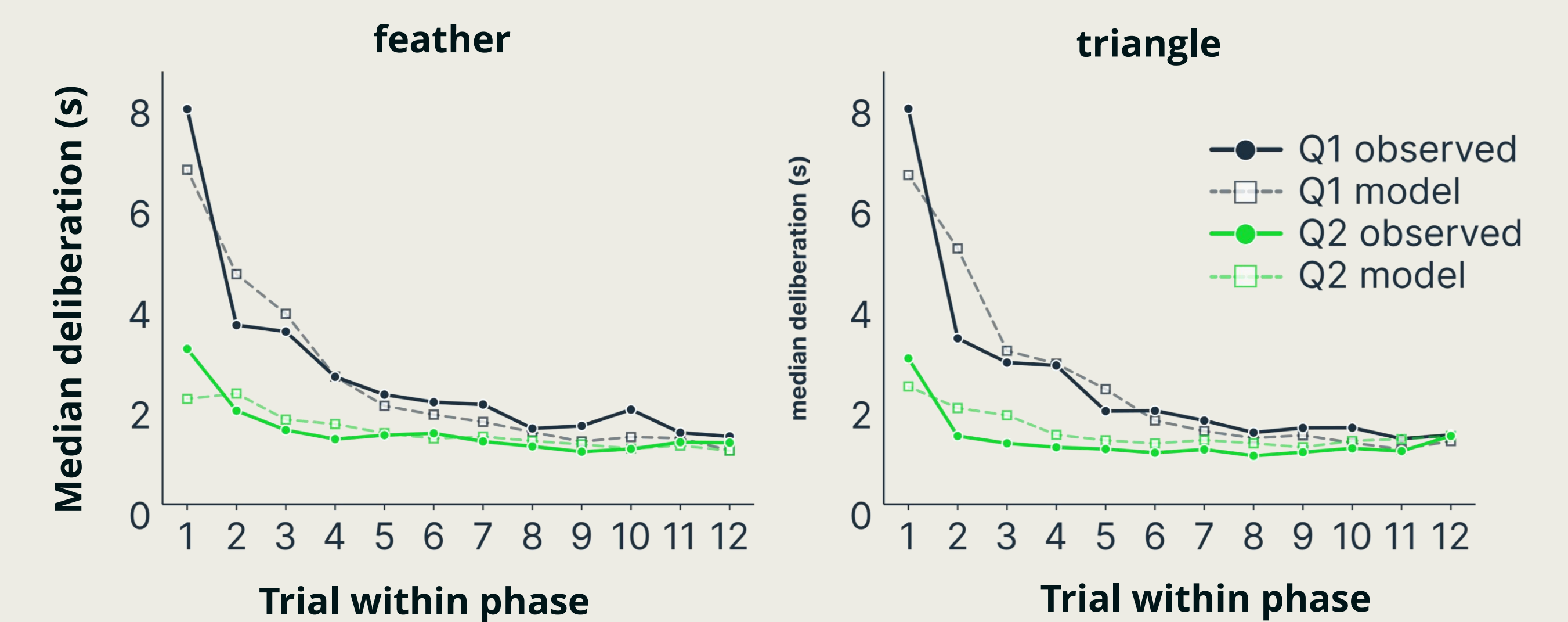
$$RT(t) = t_0 \left[e^{-\kappa(a-1)} \right] + C + \tau \cdot n_{sim}(t) + SW \cdot switch(t)$$

orienting checking Dyna sampling switch cost

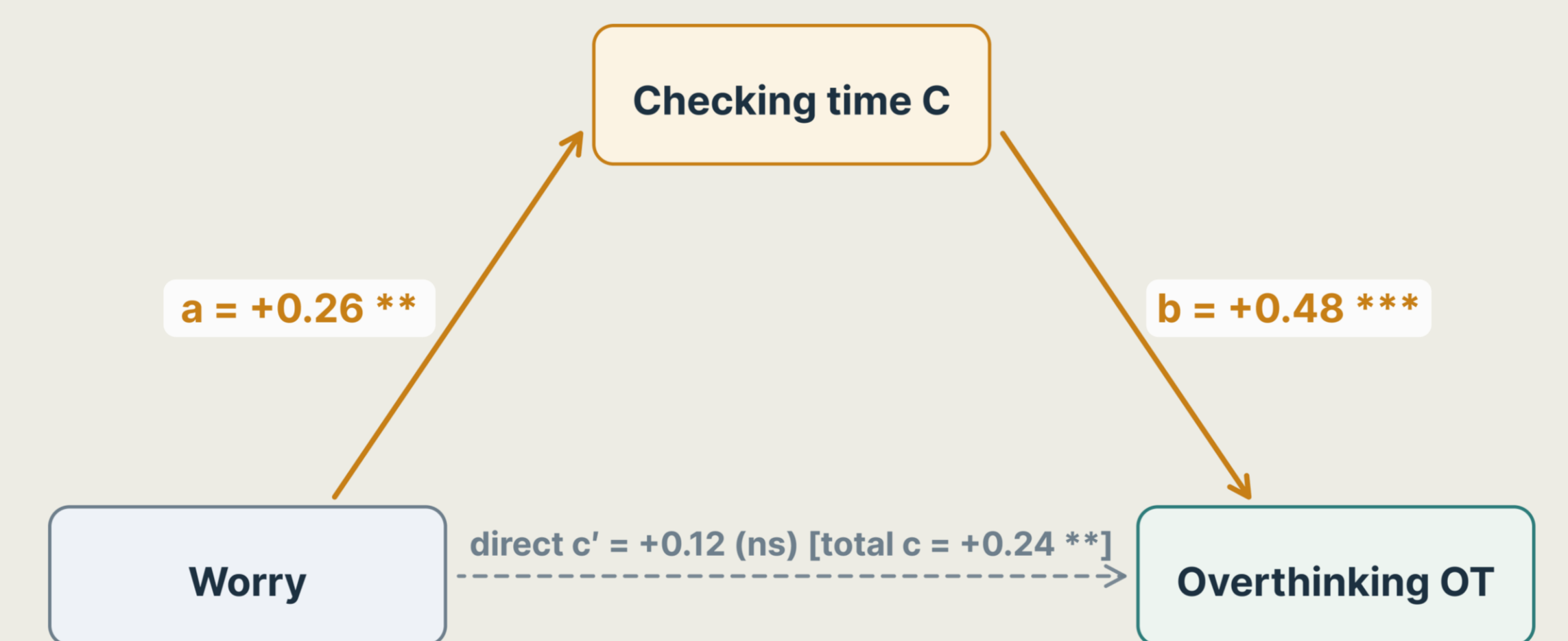
Dyna value re-valuation process across trials



Model can reproduce deliberation for both stimuli



Internal checking mediates worry -> over-thinking



Conclusion

Current modeling work suggests that worriers internally check already computed values. Future work will try to map this to subjective confidence.

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